

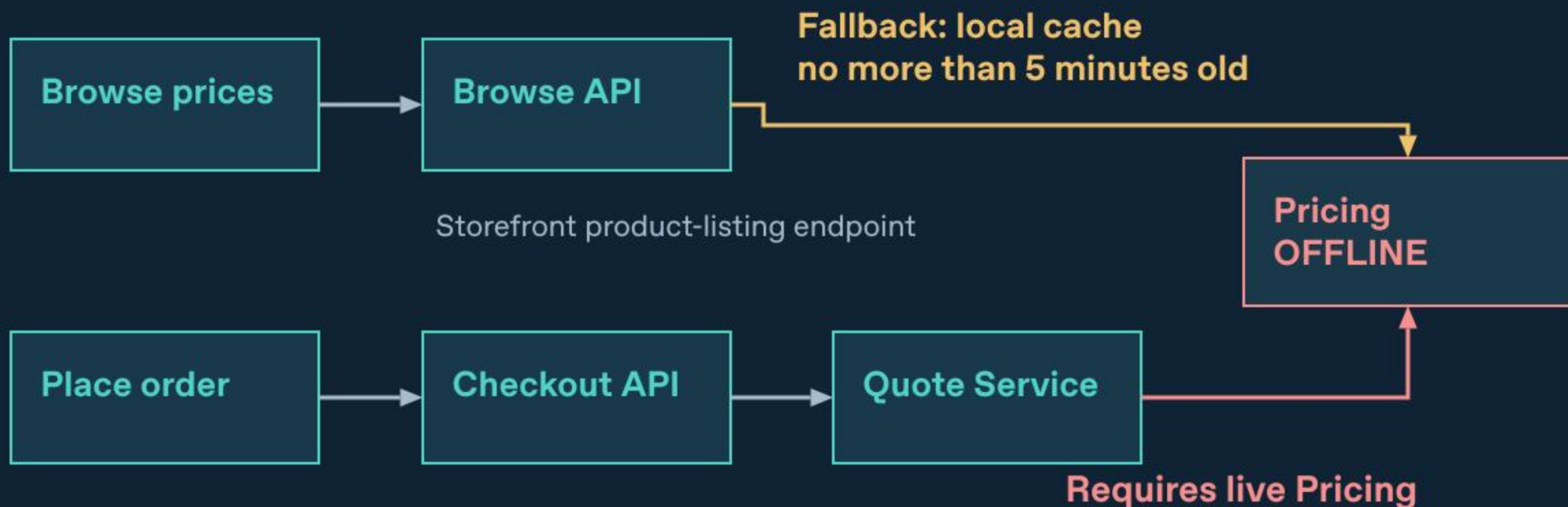
What Breaks If We Turn This Off?

From Architecture Diagram to
Dependency Graph with Corvic

Pricing maintenance: can customers still buy?

Pricing maintenance affects customers differently

Pricing is offline. All other components are available under the stated rules.



Tracking uses Orders + Shipment. It has no dependency path to Pricing.

A correct answer can still contain a wrong explanation

The diagram and extracted table



One sentence in the AI explanation



We reviewed the 15 stored relationships before reusing them.

An outage reaches customers through their dependencies



Trace back from Pricing to the journeys that depend on it

A working fallback can keep a dependency from blocking completion.

The answer no longer depends on asking AI to read the same picture again.

When the cache expires, browsing breaks too

Only the cache age changes. Pricing remains offline.

Customer journey	Cache: 2 minutes	Cache: 8 minutes
Browse prices	Works using cache	Cannot complete
Place order	Cannot complete	Cannot complete
Track order	Works normally	Works normally

Graph queries find paths. Explicit rules determine completion.

“Works using cache” corresponds to degraded. Actual six-row result is in the appendix.

One written rule explains the changed outcome

Browse price rule in the source

**Cached prices are usable
for at most five minutes**

2 minutes

Cache is allowed
Browse completes

8 minutes

Cache has expired
Browse cannot complete

Same diagram. Same stored contracts. One changed assumption.

The practical value is reusing a reviewed system model

AI reads the diagram

People review the extracted relationships and rules

Graph queries trace dependencies

The same saved relationships answer the changed scenario

Corvic keeps the work together

Source data, native queries and auditable results

Other databases and code could implement the same rules.

This model says checkout still needs Pricing

Checkout breaks at both tested cache ages

Browsing survives only while its cache is valid

Tracking stays unaffected

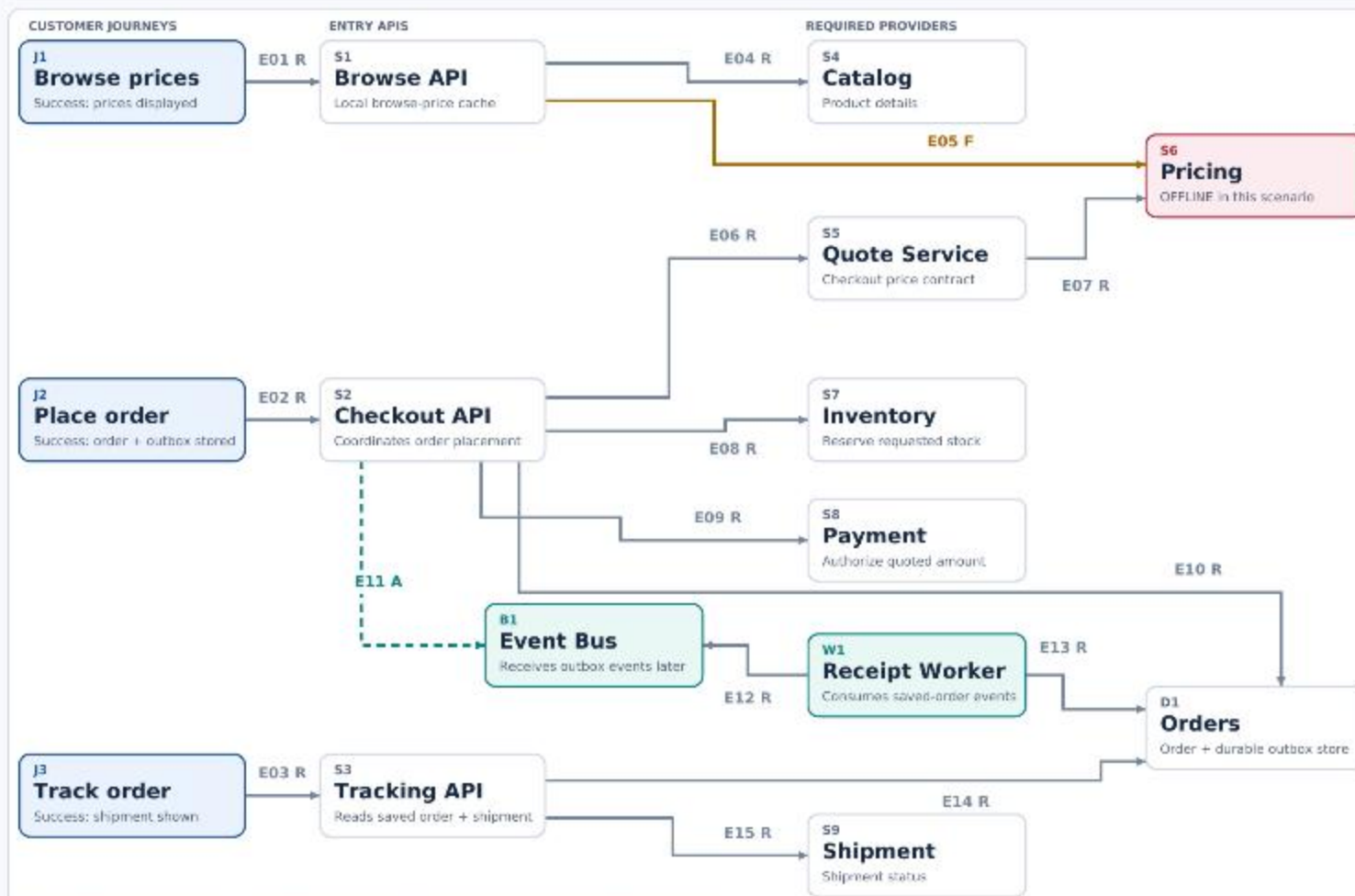
What live facts would you verify before approving maintenance?

Can we take Pricing offline?

Three customer journeys. Directed contracts. One explicitly bounded maintenance scenario.

SCENARIO Pricing is OFFLINE. Every other component is UP.

Browse cache age = 2 min (limit 5 min). Existing signed quote = valid for this basket. No hidden dependencies.



All nodes are synthetic. Component UP status is not the same as request readiness.

READ THE CONTRACTS

Arrow direction

Caller / consumer -> dependency
An arrow does not show event flow.

R REQUIRED / BLOCKING

Source requires target readiness.
All R and F contracts must hold;
follow them transitively.

F CONDITIONAL FALLBACK

Use the target if it can complete.
Otherwise use the stated local
fallback if its condition holds,
if neither is available, source fails.

A ASYNC / NONBLOCKING

Does not gate customer completion.
E11 delivers the stored outbox later.

E05 BROWSE PRICE RULE

Pricing OR local price cache.
Cache is usable iff age $\leq$ 5 min.
Cached prices satisfy Browse success.

E07 QUOTE PRICE RULE

A: Quote Service requires live Pricing.
An existing quote is NOT a fallback.

SCOPE / SUCCESS

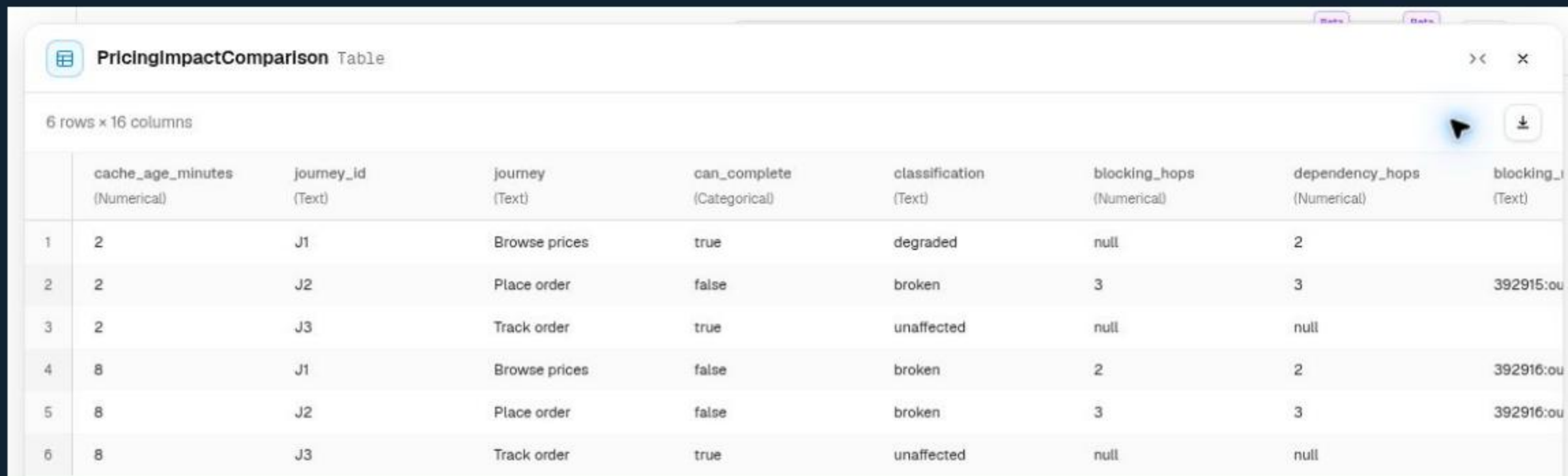
Evaluate one request per journey.
Order + outbox commit atomically.
Receipt delivery is not order success.
No retries, extra fallbacks or capacity
effects may be inferred. Crossed lines
are not joins. Use only drawn arrows.

QUESTION

Which journeys complete, which break, and why?

Cite edge IDs and directed paths. Say whether completion uses a fallback. Do not guess production behavior.

Appendix: actual computed Corvic result



	cache_age_minutes (Numerical)	journey_id (Text)	journey (Text)	can_complete (Categorical)	classification (Text)	blocking_hops (Numerical)	dependency_hops (Numerical)	blocking_ (Text)
1	2	J1	Browse prices	true	degraded	null	2	
2	2	J2	Place order	false	broken	3	3	392915:ou
3	2	J3	Track order	true	unaffected	null	null	
4	8	J1	Browse prices	false	broken	2	2	392916:ou
5	8	J2	Place order	false	broken	3	3	392916:ou
6	8	J3	Track order	true	unaffected	null	null	

Native graph reachability + deterministic SQL

Saved output 392919:output. See the runbook for view-only replay.

Appendix: direction controls and result lineage

Native direction test	Result	Native output
Quote Service, 1 hop	Pricing	392885:output_1
Pricing, 1 hop	No nodes	392885:output_2
Place order, 3 hops	Pricing	392886:output

Browse at 8 minutes: 392919 → 392916:output_2

Blocking projection 392904 → original E05 in 392839

These refs link the final classification to native queries and the stored source.

Appendix: rules, audit and limits

Explicit rules

Required dependencies block. The cache fallback has a 5-minute limit. Async receipts do not gate order completion.

Audit result

All 15 source contracts preserved. Only the browse fallback changes blocking state at 2 versus 8 minutes.

Scope

One synthetic system. Other SQL and graph tools could implement this. No comparative benchmark.